PriceSense

Demand Forecasting & Dynamic Pricing Decision Engine

PROJECT BUSINESS REPORT

Nitin

Tools: PostgreSQL | SQL | Python | Pandas | Jupyter Notebook | scikit-learn | Power BI | DAX

Dataset: Svizor Retail Sales Forecasting Data

Project Type: Demand Forecasting, Price Elasticity Analysis & Dynamic Pricing Decision Support

1. 1. Executive Summary

1.1 Project Overview

PriceSense is a data-driven Demand Forecasting and Dynamic Pricing Decision Engine designed to analyse historical retail sales, understand demand patterns, evaluate price sensitivity, and generate pricing recommendations. The project combines SQL-based data preparation, Python-based forecasting and pricing analysis, and Power BI visualisation to transform retail data into actionable business insights.

Using the Svizor Retail Sales Forecasting dataset, the project follows an end-to-end analytical workflow covering data understanding, database organisation, feature engineering, demand forecasting, price elasticity analysis, pricing simulation, and business intelligence reporting. The primary objective is to support more informed demand planning and pricing decisions through quantitative analysis.

1.2 Key Results and Deliverables

The project produced a structured PostgreSQL database, analytical datasets, a demand forecasting model, pricing simulation outputs, and an interactive Power BI dashboard.

Key deliverables include:

- **Data Engineering and SQL Analysis:** Organised retail data into raw, staging, and analytics layers for structured analysis and transformation.
- **Demand Forecasting:** Developed and evaluated a Random Forest model to predict retail demand, achieving a test (R^2) score of approximately **0.911**.
- **Price Elasticity and Simulation:** Analysed price sensitivity and simulated alternative pricing scenarios to estimate their potential effects on demand and revenue.
- **Pricing Recommendations:** Generated data-driven price adjustment recommendations for the available recommendation period.
- **Power BI Dashboard:** Developed an interactive, four-page dashboard covering executive KPIs, demand analysis, pricing and elasticity, and pricing recommendations with model performance.

The pricing outcomes represent simulation-based estimates rather than verified real-world revenue improvements.

1.3 Technology Stack

The project uses the following technologies:

- **PostgreSQL and SQL:** Database management, data transformation, analytical queries, and preparation of modelling datasets.
- **Python, Pandas and scikit-learn:** Data processing, feature engineering, demand forecasting, model evaluation, and pricing analysis.
- **Jupyter Notebook:** Exploratory analysis, experimentation, and model development.

- **Power BI and DAX:** Interactive dashboards, business KPIs, trend indicators, and analytical reporting.

Together, these tools demonstrate an end-to-end approach to retail analytics, combining data preparation, predictive modelling, pricing decision support, and business intelligence.

2. Business Problem & Project Objectives

2.1 Business Context and Problem Statement

Retail businesses must balance product demand, selling prices, discounts, and revenue to make effective commercial decisions. Historical sales data contains valuable information about customer demand, seasonal patterns, promotional activity, and price changes. However, analysing large volumes of retail data and translating these patterns into practical pricing decisions can be challenging.

PriceSense addresses this challenge by integrating historical sales analysis, demand forecasting, price elasticity analysis, and pricing simulations into a unified analytical workflow. The project aims to help businesses understand demand behaviour, estimate the potential impact of price adjustments, and make more informed pricing decisions using data-driven insights.

2.2 Project Objectives

The primary objective of PriceSense is to develop an analytical decision-support system that combines demand forecasting with dynamic pricing analysis.

The key objectives are:

- **Data Preparation:** Organise, clean, and transform retail datasets using PostgreSQL and Python.
- **Demand Analysis:** Identify historical sales patterns, demand trends, and relevant product, store, calendar, and promotional factors.
- **Demand Forecasting:** Develop and evaluate a machine learning model to estimate future demand.
- **Price Elasticity Analysis:** Examine the relationship between selling prices and product demand.
- **Pricing Simulation:** Evaluate alternative price scenarios and estimate their potential effects on demand and revenue.
- **Recommendation Generation:** Produce pricing recommendations based on simulated outcomes and the project's decision logic.
- **Business Intelligence:** Build an interactive Power BI dashboard to communicate key performance indicators, analytical findings, and model results.

2.3 Key Business Questions

PriceSense is designed to investigate the following business questions:

- How do sales and demand vary across products, stores, and time periods?
- Which historical patterns and features are useful for forecasting demand?
- How does price sensitivity influence expected product demand?
- What potential changes in demand and revenue could result from alternative pricing scenarios?
- Which price adjustments are suggested by the pricing simulation?
- How can forecasting and pricing results be presented clearly to support business decision-making?

These questions connect the technical analysis to practical retail planning and pricing decisions.

2.4 End-to-End Analytical Workflow

The project follows a structured workflow that transforms raw retail data into analytical outputs and business insights.

1. **Data Understanding:** Inspect the source datasets, their structure, data volume, and quality.
2. **Database Preparation:** Organise data in PostgreSQL using raw, staging, and analytics layers.
3. **SQL Analysis and Data Transformation:** Prepare analytical datasets and perform structured retail analysis.
4. **Feature Engineering and EDA:** Develop relevant demand, calendar, pricing, product, store, and promotional features and investigate historical patterns.
5. **Demand Forecasting:** Train and evaluate machine learning models for demand prediction.
6. **Price Elasticity Analysis:** Examine price sensitivity to support pricing simulations.
7. **Dynamic Pricing Simulation:** Compare candidate price adjustments and estimate their potential demand and revenue effects.
8. **Power BI Reporting:** Present KPIs, demand trends, pricing insights, recommendations, and model performance through an interactive dashboard.

This workflow brings together data engineering, predictive analytics, pricing decision support, and business intelligence within a single project.

3. Dataset & Data Understanding

3.1 Dataset Overview and Source

PriceSense uses the **Svizor Retail Sales Forecasting Data** dataset obtained from Kaggle. The dataset contains historical retail information covering multiple stores, products, sales transactions, pricing, discounts, markdowns, and related product and store attributes.

The available data provides a foundation for studying historical demand patterns, understanding price behaviour, developing forecasting features, and evaluating alternative pricing scenarios. The recorded price-history period extends from **28 August 2022 to 26 September 2024**.

3.2 Source Files and Business Purpose

The project uses eight original CSV files, each contributing a different aspect of the retail analysis.

Source file	Purpose
sales.csv	Historical sales and demand information
catalog.csv	Product catalogue and product attributes
stores.csv	Store-level information
price_history.csv	Historical product pricing information
discounts_history.csv	Historical discount-related information
markdowns.csv	Markdown and promotional price-reduction information
online.csv	Online retail-related information
actual_matrix.csv	Actual-value matrix supplied with the dataset

These files were examined to understand their structure, business relevance, and potential contribution to the analytical workflow. Their information was subsequently organised for data preparation, feature engineering, forecasting, and pricing analysis.

3.3 Data Volume, Grain and Key Identifiers

The dataset contains millions of records across its source files. During initial profiling, sales.csv was found to contain approximately **7.43 million rows**, making efficient data handling and structured database processing important parts of the project.

The core analytical grain is based on **date, item ID, and store ID**. This combination supports demand analysis at the product-store-date level and helps align sales information with relevant pricing and contextual features.

The main identifiers used throughout the analysis include:

- **Date:** Establishes the time period for historical analysis and forecasting.
- **Item ID:** Identifies individual products.
- **Store ID:** Distinguishes retail locations.

Maintaining a consistent analytical grain is essential for combining datasets and constructing meaningful demand-related features.

3.4 Data Profiling and Quality Assessment

Initial data profiling was performed to examine record counts, column structures, missing values, unique identifiers, date coverage, and other data characteristics. This helped identify data quality considerations before the datasets were used for further analysis.

The profiling highlighted missing values in several product attribute and promotion-related fields, including weight, fatness, promotion, and discount percentage information. These findings indicated the need to assess missingness carefully and apply appropriate handling during data preparation.

The profiling stage also helped establish the dataset's scale, identify relevant analytical fields, and understand the challenges associated with processing large retail datasets in PostgreSQL and Python.

3.5 Data Coverage and Limitations

Although the dataset supports historical demand forecasting and pricing analysis, its coverage and quality impose certain limitations. Missing attributes may restrict some product-level comparisons, while historical relationships between price and demand do not necessarily establish direct causation.

Additionally, the pricing recommendations and simulations cover a more limited period than the complete historical dataset. Consequently, the recommendations should be interpreted within their available date range rather than as pricing decisions for the entire historical period.

Understanding these limitations provides important context for interpreting the forecasting results, simulated revenue effects, and business insights presented in the subsequent sections.

4. PostgreSQL Database Architecture & SQL Analysis

4.1 Database Design and Schema Organisation

PostgreSQL was used as the central database for organising and analysing the retail dataset. The database, named pricesense, was structured to support data preparation, analytical transformations, and the creation of datasets required for demand forecasting and dynamic pricing.

The database approach provided a structured environment for working with large volumes of retail data while separating the original source data from processed and analysis-ready tables.

4.2 Raw, Staging and Analytics Layers

The database was organised into three logical schemas: raw, staging, and analytics. Each schema served a distinct purpose within the data preparation workflow.

- **Raw Layer:** Stored the original dataset tables corresponding to the eight source CSV files, preserving the source information for reference.
- **Staging Layer:** Provided a separate layer for data preparation, cleaning, and intermediate transformations.
- **Analytics Layer:** Contained derived features, analytical datasets, model outputs, and pricing-related results used in subsequent analysis and reporting.

This layered structure improved the organisation of the database and separated source data from transformed data and analytical outputs.

4.3 SQL-Based Data Preparation and Transformations

SQL was used in PostgreSQL to inspect datasets, examine data distributions, analyse records, and prepare information for downstream analytical tasks. Queries supported data validation, feature preparation, and the organisation of sales, product, store, pricing, and promotional information.

Analytical tables were developed to bring together relevant information and support demand forecasting and pricing analysis. This approach also reduced the need to repeatedly work directly with the original large source files during subsequent stages of the project.

4.4 Historical Sales, Demand and Pricing Analysis

SQL analysis focused on understanding historical sales behaviour, demand patterns, product and store characteristics, and pricing-related information. These analyses provided the foundation for identifying relevant variables and preparing data for exploratory analysis, model development, and pricing simulations.

Sales-derived and price-related analytical tables supported the integration of historical demand information with relevant features. This helped establish a consistent data foundation for the forecasting and pricing components of PriceSense.

4.5 Data Validation and Analytical Table Design

Data validation and structured table design were important for maintaining consistency throughout the analytical workflow. The preparation process considered record counts, data types, date coverage, missing values, key identifiers, and the alignment of data at the appropriate analytical grain.

The resulting analytics layer included feature tables such as analytics.sales_derived_features, analytics.sales_markdown_features, and analytics.price_sense_features, alongside model and pricing outputs such as analytics.model_comparison, analytics.pricing_recommendations, and analytics.pricing_summary.

Overall, PostgreSQL provided the data foundation for PriceSense, connecting the original retail datasets with feature engineering, forecasting, pricing analysis, and Power BI reporting.

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5. 5. Feature Engineering & Exploratory Data Analysis

5.1 Calendar and Time-Based Features

Calendar information is important in retail demand analysis because sales can vary across different dates and time periods. PriceSense incorporated date-based information into the analytical workflow to support historical demand analysis and forecasting.

The project maintained the dataset's historical date coverage from 28 August 2022 to 26 September 2024. Time-based features helped organise observations chronologically and provided temporal context for analysing demand patterns.

5.2 Demand Lags and Rolling Statistics

Historical demand can provide valuable information about future demand. PriceSense incorporated lag-based features and rolling demand statistics to represent previous sales behaviour.

The engineered demand features included:

- **Lag features:** Previous demand observations at intervals of 1, 7, 14, and 28 days.
- **Rolling statistics:** Aggregated historical demand measures used to capture broader demand patterns.

These features were designed to help the forecasting model identify recent demand behaviour and recurring patterns. Their availability depends on the amount of preceding historical data, so initial observations may not have values for every lag or rolling feature.

5.3 Price, Product, Store and Promotion Features

The project considered multiple categories of information to support demand and pricing analysis. These included historical prices, product attributes, store information, discount-related data, and markdown or promotional information.

Relevant data was organised through analytical feature tables, including analytics.sales_derived_features, analytics.sales_markdown_features, and analytics.price_sense_features. These datasets provided a foundation for examining the relationship between demand and its potential explanatory factors.

5.4 Demand and Sales Pattern Analysis

Exploratory data analysis focused on understanding historical sales behaviour and variation across products, stores, and time periods. Examining these patterns helped establish the context for demand forecasting and identify relevant information for feature engineering.

The analysis also supported the identification of temporal patterns and differences in demand behaviour that could influence forecasting performance. These observations informed the subsequent model development and evaluation stages.

5.5 Price, Discount and Markdown Analysis

Pricing-related exploratory analysis examined historical selling prices, discounts, and markdown information to understand how retail pricing varied over time.

This stage provided the groundwork for the project's price elasticity analysis and pricing simulations. However, observed changes in sales alongside price changes do not independently establish causation, as promotional activity, seasonality, and other factors may also influence demand.

5.6 Key Exploratory Findings

The profiling and exploratory stages established several important considerations for the subsequent analysis:

- The dataset contained millions of sales records, requiring structured data processing.
- Historical demand features could be used to represent recent and longer-term sales behaviour.
- Product, store, pricing, and promotional information provided additional analytical context.
- Missing values in certain product and promotion-related fields required consideration during data preparation.
- Historical price and demand relationships provided a starting point for elasticity analysis and pricing simulations, while requiring careful interpretation.

Overall, feature engineering and exploratory analysis transformed the initial understanding of the retail dataset into a more structured foundation for demand forecasting and dynamic pricing decision support.

6. Demand Forecasting & Model Development

6.1 Forecasting Objective and Target Definition

The objective of demand forecasting in PriceSense was to estimate retail demand using historical sales information and relevant explanatory features. Accurate demand estimates can support inventory planning, demand analysis, and subsequent pricing simulations.

The forecasting workflow used historical demand observations as the target variable and incorporated relevant engineered features to capture previous sales behaviour and other contextual information. Demand forecasting formed the predictive component of the broader PriceSense decision-support system.

6.2 Modelling Dataset and Feature Preparation

The modelling dataset was prepared using the analytical data developed during the data preparation and feature engineering stages. Relevant features included historical demand lags, rolling demand statistics, and other available contextual variables.

Lag features representing demand from 1, 7, 14, and 28 days earlier were included in the feature engineering workflow. These features were intended to help the model learn relationships between previous demand behaviour and the demand being predicted.

The modelling process also required chronological separation of training, validation, and testing data to evaluate performance on observations outside the training data. Historical features must be constructed carefully to avoid using information that would not have been available at prediction time.

6.3 Model Selection and Comparison

PriceSense included a model comparison stage to evaluate candidate forecasting approaches before selecting the model used in the forecasting workflow. Model performance was assessed using quantitative evaluation metrics rather than relying solely on visual inspection of predictions.

The Random Forest model was used for the final forecasting approach. Its ability to learn nonlinear relationships made it suitable for exploring demand patterns across the available historical features.

6.4 Random Forest Forecasting Approach

Random Forest is an ensemble machine learning algorithm that combines predictions from multiple decision trees to produce a final estimate. For regression tasks, the model aggregates the predictions of individual trees to generate a continuous numerical output.

In PriceSense, Random Forest was used to estimate demand from the prepared modelling features. The resulting forecasts provided predicted demand values for the available forecast period and served as an input to the wider pricing decision-support workflow.

6.5 Training, Validation and Testing

The model evaluation process used training, validation, and testing stages to assess forecasting performance. Training data was used to fit the model, validation data supported performance assessment during model development, and test data provided an additional evaluation of predictive accuracy.The use of separate evaluation datasets helped assess how well the model performed on data that was not used to fit it. For time-dependent retail data, chronological separation is particularly important because random splitting can allow future information to influence the evaluation.

6.6 Forecasting Metrics and Results

The Random Forest model was evaluated using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the coefficient of determination ((R^2)). MAE measures the average absolute difference between predicted and actual demand, while RMSE gives greater weight to larger prediction errors. The \(R^2\) metric measures how well the model's predictions explain variations in the actual demand values.

Evaluation metric	Validation result	Test result
MAE	2.218	2.145
RMSE	7.132	9.701
(R^2)	0.929	0.911

The validation (R^2) score of approximately 0.929 and test (R^2) score of approximately 0.911 indicate that the model explained a substantial proportion of the variation in the target values within the respective evaluation datasets.

6.7 Model Performance Interpretation

The Random Forest model achieved a test MAE of approximately 2.145 and a test RMSE of approximately 9.701. MAE represents the average absolute prediction error, while RMSE penalises larger errors more heavily. Both metrics are expressed in the units of the forecasting target.

The difference between MAE and RMSE suggests that larger prediction errors may have contributed disproportionately to the overall test error. Although the (R^2) results indicate strong predictive performance on the evaluated data, these metrics alone do not establish that the model will perform equally well across every product, store, or future period.

The forecasts should therefore be interpreted alongside the evaluation methodology, data coverage, and limitations of the available dataset. Within PriceSense, the forecasting model provides a quantitative foundation for demand analysis and subsequent pricing simulations.

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7. Price Elasticity Analysis

7.1 Pricing and Demand Relationship

Price elasticity analysis helps businesses understand how demand may respond to changes in selling prices. In retail, the relationship between price and demand can vary across products and may also be influenced by promotions, seasonality, and other factors.

Within PriceSense, historical sales and pricing information provided the foundation for examining price sensitivity. This analysis formed an important link between demand forecasting and the dynamic pricing simulation process.

7.2 Price Elasticity Methodology

Price elasticity of demand measures the percentage change in quantity demanded associated with a percentage change in price. It provides a way to represent the sensitivity of demand to price adjustments.

PriceSense used elasticity values in its pricing simulation logic to estimate how demand might change under alternative price scenarios. The general relationship was represented using the following formula:

Simulated Demand = Current Demand × (Simulated Price / Current Price) ^ Price Elasticity

Where:

- Simulated Demand: Estimated demand at the proposed price.
- Current Demand: Baseline demand before the price adjustment.
- Simulated Price: The alternative price being evaluated.
- Current Price: The existing selling price.
- Price Elasticity: The elasticity value used to estimate the demand response to a price change.

This relationship allows the engine to estimate demand at alternative price levels and calculate the corresponding simulated revenue.

7.3 Elasticity Interpretation

Price elasticity helps explain the potential direction and magnitude of demand responses to price changes. When elasticity is negative, an increase in price generally corresponds to a decrease in estimated demand, while a price reduction generally corresponds to an increase in estimated demand, assuming other factors remain unchanged.

The magnitude of elasticity indicates the sensitivity of demand to price changes. However, the resulting estimates depend on the reliability and suitability of the elasticity values used in the simulation.

Within PriceSense, elasticity analysis supports the comparison of alternative pricing scenarios rather than guaranteeing a particular demand or revenue outcome.

7.4 Assumptions and Analytical Limitations

The pricing simulation assumes that the selected elasticity value provides a reasonable representation of demand sensitivity over the price range being evaluated. It also assumes that other relevant factors remain sufficiently stable for the estimated relationship to be useful.

In practice, demand may also respond to competitor prices, promotions, seasonality, product availability, and customer behaviour. Historical price and sales relationships may not fully isolate the effect of price changes from these influences.

Therefore, the elasticity-based outputs should be interpreted as **simulation estimates for decision support**, not as proof of causal relationships or guaranteed business results. Further validation using controlled pricing experiments or additional business data could improve the reliability of future recommendations.

8. Dynamic Pricing Simulation & Recommendation Engine

8.1 Baseline Price and Demand

The pricing simulation uses the current selling price and baseline demand as reference values for evaluating alternative pricing decisions. These values provide a starting point for estimating how demand and revenue may change when a product's price is adjusted.

Within PriceSense, historical pricing information and demand-related analytical outputs support the evaluation of candidate price changes over the available recommendation period.

8.2 Candidate Pricing Scenarios

PriceSense evaluates alternative pricing scenarios by considering price adjustments relative to the current selling price. The scenarios allow potential price increases and decreases to be compared using simulated demand and revenue outcomes.

The generated pricing recommendations include price decreases of 10% and price increases of 10%. These adjustments provide a consistent basis for examining the potential effects of alternative pricing decisions.

8.3 Simulated Demand and Revenue Calculations

The engine estimates demand under each candidate price using the price elasticity relationship established in the previous section. The resulting simulated demand is then used to calculate potential revenue.

Simulated Revenue = Simulated Price × Simulated Demand

By applying these calculations to alternative price scenarios, PriceSense estimates how changes in selling prices may affect demand and revenue. These outputs represent model-based estimates rather than confirmed real-world outcomes.

8.4 Pricing Decision and Recommendation Logic

The recommendation engine uses the pricing simulation outputs to support price adjustment decisions. Candidate price changes are evaluated using the estimated demand and revenue associated with each scenario.

The resulting recommendations provide a structured way to identify potential price increases or decreases. However, the recommendations should be treated as decision-support outputs and considered alongside business constraints, customer behaviour, and other commercial factors before implementation.

8.5 Scenario Comparison and Recommendation Outputs

The project stores pricing-related outputs in PostgreSQL, including analytics.pricing_recommendations, analytics.pricing_scenarios, and analytics.pricing_summary.

These outputs support the examination of proposed price adjustments, simulated outcomes, and summary-level pricing results. The recommendations cover the period from **1 July 2024 to 26 September 2024**, which is more limited than the complete historical dataset.

8.6 Estimated Revenue Impact and Interpretation

The simulated pricing scenarios provide an estimate of the potential revenue implications of alternative price adjustments. Comparing these estimates helps illustrate how price sensitivity and proposed selling prices can influence projected commercial outcomes.However, simulated revenue uplift should not be interpreted as actual revenue growth. The estimates depend on the assumptions used in the demand and elasticity calculations, and real-world outcomes may differ because of promotions, competitor actions, inventory availability, and customer responses.

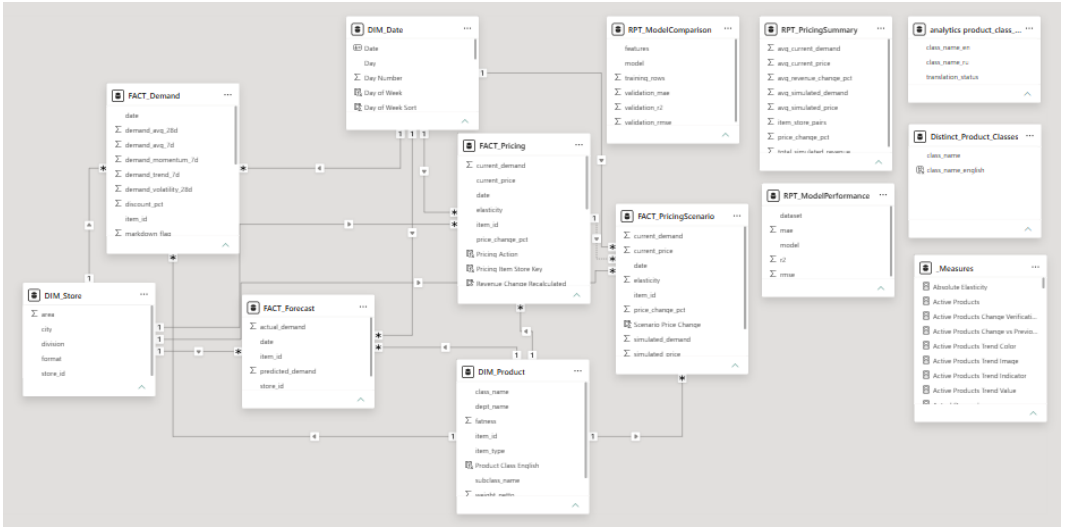
Overall, the pricing recommendation engine connects demand forecasting and elasticity analysis with a structured approach to evaluating alternative pricing decisions.

9. Power BI Data Model & KPI Framework

9.1 Data Model and Relationships

Power BI was used to connect the analytical outputs of PriceSense with an interactive reporting layer. The data model supports the analysis of historical demand, revenue, product performance, pricing scenarios, recommendations, and model evaluation results.

The model was designed to provide a consistent foundation for dashboard calculations and visualisations. Appropriate relationships between analytical tables and shared dimensions are important for ensuring that filters and measures return meaningful results.



9.2 Date Table and Time-Based Analysis

A dedicated date dimension, DIM_Date, supports time-based analysis throughout the dashboard. It enables users to filter data by date and month and examine changes across selected periods.

The reporting model preserves the historical dataset's date coverage from 28 August 2022 to 26 September 2024. Date filtering is particularly important when analysing historical sales trends and when evaluating pricing recommendations, which are available only for the period from July to September 2024.

9.3 DAX Measures and KPI Definitions

Data Analysis Expressions (DAX) was used to create measures for calculating business KPIs and presenting analytical results dynamically. These measures respond to the selected filter context, allowing users to investigate different time periods and aspects of retail performance.

The Executive Overview includes six key performance indicators:

- Total Revenue
- Net Units Sold
- Active Products
- Average Selling Price
- Recommended Price Change %
- Recommended Revenue Change %

Additional measures support demand analysis, pricing and elasticity analysis, revenue simulation, and model performance reporting.

9.4 Trend Indicators and Conditional Formatting

Trend measures were developed to communicate changes in key metrics through formatted percentages, directional indicators, and conditional colours. Positive changes are represented in teal, negative changes in red, and unavailable or neutral comparisons in grey.

These visual indicators help users interpret changes more quickly. Where a meaningful comparison cannot be calculated, the dashboard uses a neutral representation rather than implying a positive or negative result.

9.5 Filter Behaviour and Measure Validation

The dashboard's measures were examined under different date selections and filter conditions to improve the reliability of the displayed results. Particular attention was given to period comparisons, date boundaries, and the handling of blank values.

Historical trend comparisons require an appropriate previous period, while pricing recommendation measures must respect the limited period for which recommendation data exists. These considerations are essential for preventing misleading KPI values and ensuring that the dashboard communicates results within the correct analytical context.

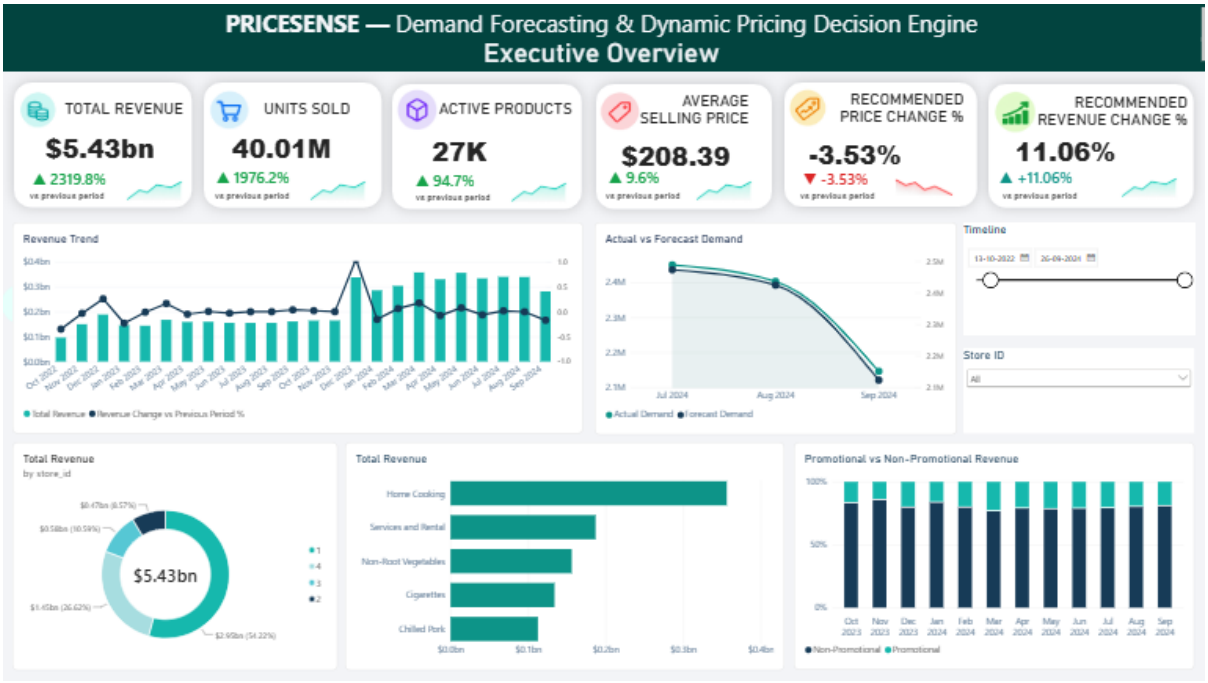
Overall, the Power BI data model and DAX framework connect the project's analytical outputs with interactive KPIs, trend indicators, and business reporting.

10. Power BI Dashboard & Business Insights

10.1 Executive Overview

The Executive Overview provides a high-level summary of retail performance and pricing recommendations. It presents six KPI cards covering Total Revenue, Net Units Sold, Active Products, Average Selling Price, Recommended Price Change %, and Recommended Revenue Change %.

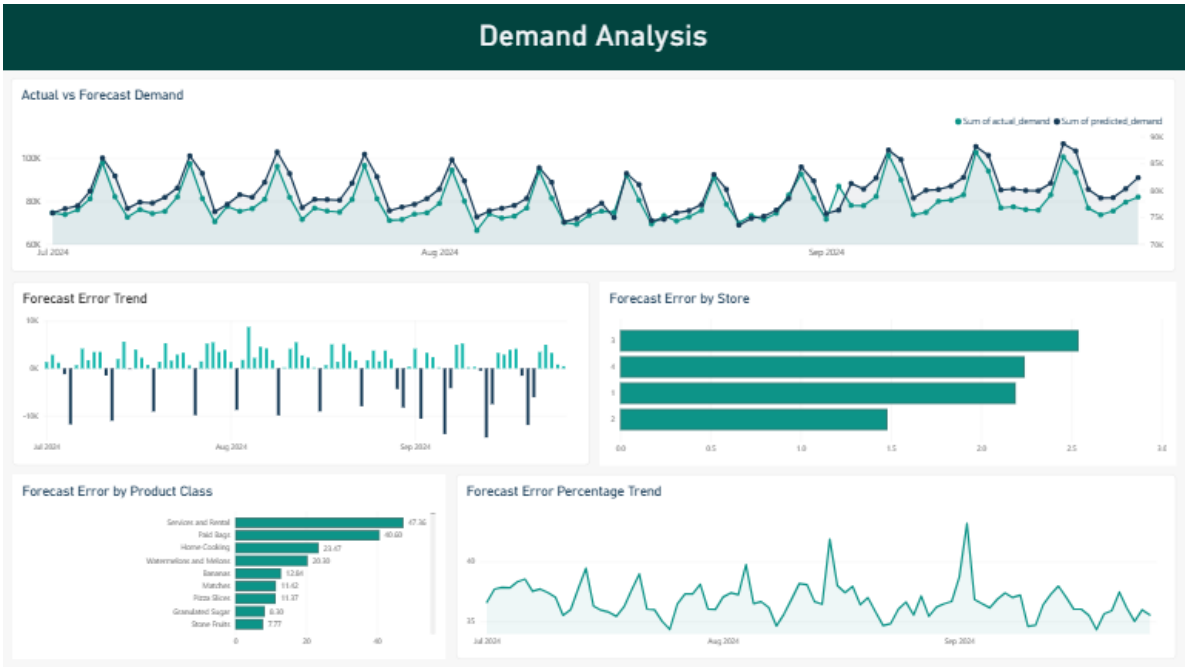
The page is designed to help users quickly assess overall business performance and understand the potential impact of proposed pricing adjustments. KPI cards, trend indicators, and supporting visuals provide a concise starting point for further analysis.



10.2 Demand Analysis

The Demand Analysis page focuses on historical sales and demand behaviour across the available products, stores, and time periods. It helps users examine demand patterns and understand how performance varies over time.

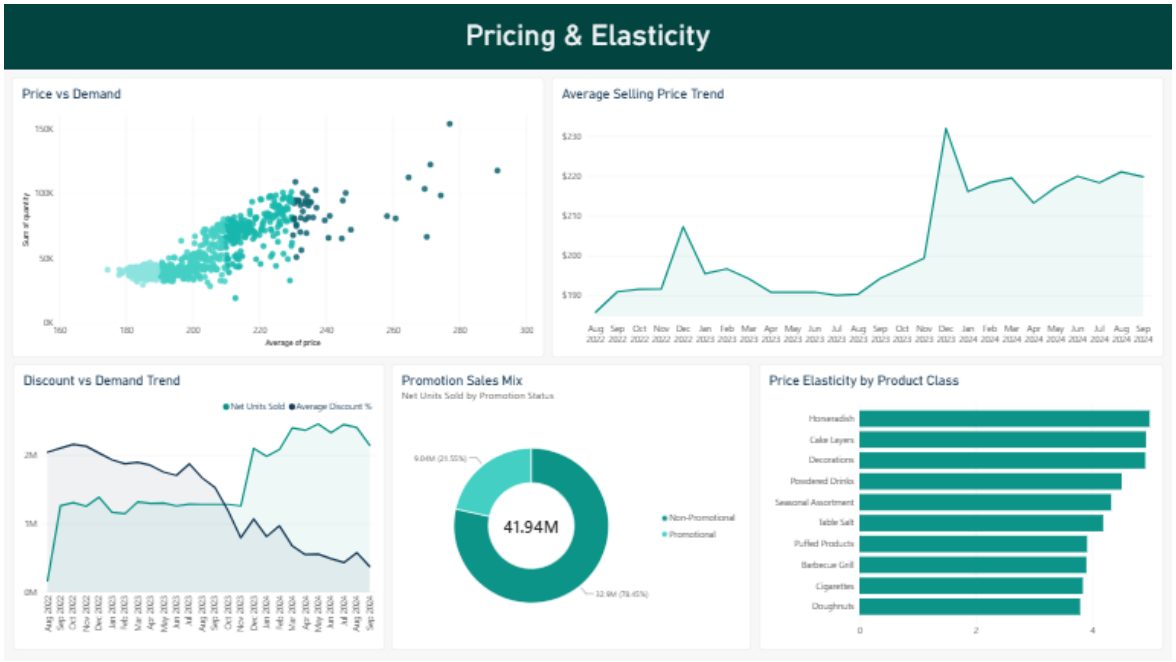
The page supports the interpretation of historical demand information and provides context for the demand forecasting component of PriceSense. Its purpose is to make demand-related patterns easier to explore through interactive visualisations and filters.



10.3 Pricing & Elasticity

The Pricing & Elasticity page focuses on historical pricing information and the relationship between selling prices and demand. It supports the examination of price sensitivity and provides context for the assumptions used in the pricing simulation.

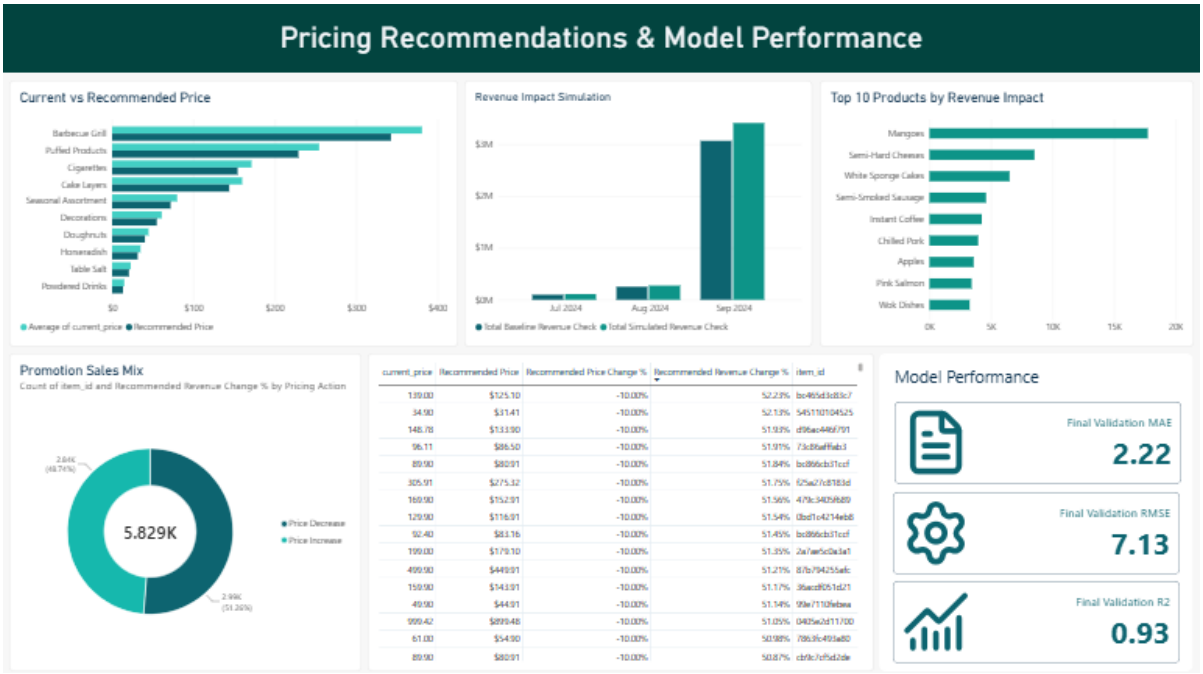
By bringing pricing-related information into a dedicated reporting page, the dashboard helps users explore the potential implications of price adjustments and understand the analytical foundation of the recommendation engine.



10.4 Pricing Recommendations & Model Performance

This page brings together the generated pricing recommendations and the forecasting model's evaluation results. It enables users to examine proposed price adjustments alongside the simulated outcomes and the model metrics used to assess predictive performance.

The recommendations cover the period from 1 July 2024 to 26 September 2024. Keeping this reporting period clear is important because the recommendations do not span the entire historical dataset.



10.5 Dashboard Design and Interactivity

The dashboard follows a consistent visual theme using a pale blue-white background, navy headings, teal accents, and white KPI cards. This design creates a clean reporting experience while maintaining a clear distinction between primary metrics and supporting visuals.

Interactive date filters, slicers, KPI cards, charts, and trend indicators allow users to explore the available data from different perspectives. Consistent formatting and appropriate handling of unavailable comparisons help improve readability and interpretation.

10.6 Key Insights Communicated Through the Dashboard

The four dashboard pages connect historical business performance with predictive and pricing analysis. Together, they allow users to explore demand behaviour, review pricing-related patterns, assess forecasting performance, and examine proposed price adjustments.

The dashboard also distinguishes historical observations from simulated pricing outcomes. This is important because forecast accuracy and simulated revenue changes provide analytical evidence for decision support, but do not independently guarantee future business results.

11. Business Recommendations, Validation & Limitations

11.1 Demand Planning and Forecast Utilisation

The demand forecasting component of PriceSense can support demand planning by providing estimates based on historical sales patterns and engineered features. Businesses could use such forecasts as an additional input when reviewing expected demand and planning inventory requirements.

However, forecasts should be assessed alongside actual sales, product availability, and changing market conditions. Regular monitoring of forecast errors would help identify areas where the model may require improvement.

11.2 Pricing Decision Support

PriceSense uses price elasticity analysis and pricing simulations to evaluate potential price adjustments. The generated recommendations can help businesses compare alternative selling prices and examine their estimated effects on demand and revenue.

Before implementing a recommendation, businesses should also consider competitor pricing, promotional activity, customer response, inventory constraints, and commercial objectives. The engine should therefore be treated as a decision-support tool rather than an automatic pricing system.

11.3 Data and Model Validation

Data profiling, analytical table preparation, and model evaluation form important parts of the PriceSense workflow. The Random Forest forecasting model was assessed using MAE, RMSE, and R^2 to measure predictive performance on validation and test data.

The analytical outputs should also be checked for data consistency, appropriate date coverage, and correct interpretation of pricing scenarios. These checks help improve confidence in the results and reduce the risk of drawing conclusions from incomplete or misleading information.

11.4 Operational Constraints and Risk Considerations

The reliability of the analysis depends on the quality and coverage of the available data, the suitability of engineered features, and the assumptions used in the forecasting and pricing processes.

Missing values, changing demand patterns, promotional effects, and differences in customer price sensitivity may influence the results. In addition, the computational requirements of processing large retail datasets highlight the importance of efficient data preparation and query design.

These considerations should be addressed before extending the project into a production environment.

11.5 Simulation Estimates Versus Realised Business Outcomes

The revenue and demand outcomes produced by PriceSense are simulation-based estimates. They indicate how alternative prices may affect demand and revenue under the assumptions used by the pricing engine.

Actual business outcomes may differ because customer behaviour and market conditions can change after a price adjustment. The project has not established that the simulated revenue effects were realised in a live retail environment.

Future validation through controlled pricing experiments, historical backtesting, or comparison with observed outcomes could help determine how reliably the recommendations translate into business performance.

12. Conclusion & Future Enhancements

12.1 Project Outcomes and Key Learnings

PriceSense demonstrates an end-to-end approach to retail analytics by combining PostgreSQL-based data preparation, Python-based demand forecasting, price elasticity analysis, pricing simulation, and Power BI reporting.

The project transforms large volumes of historical retail data into structured analytical outputs that support demand prediction and pricing decision-making. The Random Forest model achieved a test R^2 score of approximately 0.911, while the pricing simulation generated recommendations based on alternative price adjustments and estimated demand and revenue outcomes.

The project also highlights the importance of data quality, feature engineering, model evaluation, and careful interpretation of simulated results when developing data-driven business solutions.

12.2 Technical and Business Contributions

From a technical perspective, PriceSense integrates database organisation, SQL analysis, feature engineering, machine learning, pricing calculations, and interactive business intelligence into a unified workflow.

From a business perspective, it provides a framework for examining demand patterns, understanding price sensitivity, comparing potential pricing scenarios, and communicating analytical findings through a dashboard. These capabilities demonstrate how data analytics can support more structured and evidence-based retail decisions.

12.3 Future Improvements and Deployment Opportunities

Several enhancements could extend the capabilities and practical reliability of PriceSense:

- **Advanced Forecasting:** Evaluate additional forecasting models and compare their performance across different products and stores.
- **Improved Elasticity Estimation:** Refine price sensitivity estimates using more robust statistical methods and additional explanatory variables.
- **Historical Backtesting:** Evaluate pricing recommendations against historical outcomes to assess their reliability and potential business impact.
- **Business Constraint Integration:** Incorporate profit margins, inventory availability, competitor pricing, and minimum or maximum price limits into the recommendation logic.
- **Automated Data Pipeline:** Develop a repeatable process for refreshing data, generating forecasts, and updating pricing recommendations.
- **Enhanced Decision Support:** Extend the Power BI dashboard with additional validation metrics, scenario comparisons, and monitoring capabilities.

Overall, PriceSense establishes a foundation for combining predictive analytics and pricing simulation in retail decision support. With further validation and development, the project could evolve into a more comprehensive system for evaluating demand and pricing strategies.

13. Project Structure & Files

The PriceSense project is organised to separate data analysis, database scripts, model development, dashboard reporting, and project documentation. This structure improves maintainability, simplifies navigation, and makes the project easier to understand and reproduce.

Folder / File	Contents
sql/	PostgreSQL scripts organized into 01_raw/, 02_staging/, 03_analytics/, 04_model_pricing/, 05_powerbi_views/, 06_reference_translation/, and 07_validation/.
notebooks/	Jupyter notebooks for data profiling and model comparison, including 01_Data_Profiling.ipynb and 02_Model_Comparison.ipynb.
powerbi/	Power BI dashboard resources and links to the full .pbix report hosted on Google Drive.
powerbi/ss/	Screenshots of the four Power BI reporting pages: Executive Overview, Demand Analysis, Pricing & Elasticity, and Pricing Recommendations & Model Performance.
requirements.txt	Python dependencies for data analysis, machine learning, visualization, and PostgreSQL connectivity.
README.md	Project overview, objectives, workflow, technology stack, key results, and setup instructions.
report/	The final PriceSense project report in PDF format.

The project uses PostgreSQL as its database environment, with raw, staging, and analytics schemas separating source data, intermediate processing, and analytical outputs. Large original CSV files and database exports should generally remain outside the GitHub repository. The repository can instead contain the relevant scripts, notebooks, documentation, and selected outputs required to explain the project.

13.1 GitHub Repository

The GitHub repository is intended to showcase the PriceSense workflow, including its SQL analysis, forecasting approach, pricing simulation, Power BI dashboard, and supporting documentation. It should provide a clear overview of the project's objectives, methodology, key evaluation results, and limitations.

The original dataset is available through its Kaggle source, so large source files do not need to be committed to the repository. Any generated CSV outputs should be included selectively, based on their size, relevance, and reproducibility value. Database credentials, private configuration files, and unnecessary large exports should be excluded.

For reproducibility, the repository should include the SQL scripts and notebook files used in the project once they have been organised and saved. The README should also explain the required tools, dataset source, execution workflow, and any setup steps needed to reproduce the analysis.

13.2 Future Enhancements

PriceSense can be extended into a more automated and comprehensive retail pricing decision-support system through several improvements:

- **Automated Data Refresh:** Develop a repeatable pipeline to ingest new retail data and refresh analytical tables.
- **Forecast Monitoring:** Track forecast errors over time and compare actual demand with predicted demand.
- **Historical Pricing Backtesting:** Evaluate historical recommendations against observed outcomes to assess their reliability.
- **Advanced Pricing Optimisation:** Incorporate profit margins, inventory constraints, competitor prices, and business rules into pricing decisions.
- **Dynamic Power BI Analysis:** Introduce Field Parameters to let users switch between selected dimensions and measures without creating separate visuals for every combination.
- **Deployment and Monitoring:** Explore scheduled model execution and reporting updates to make the workflow more suitable for ongoing business use.

These enhancements could improve the scalability, reliability, and practical usefulness of PriceSense while building on the existing data engineering, forecasting, pricing analysis, and business intelligence workflow.